

Additive Synthesis+

*We would** like to find an efficient way of computing the sine function iteratively.

$$y(t) = \sin(2\pi ft)$$

Discretize:

$$y_n = y(t_n) = \sin\left(\frac{2\pi f_n}{R}\right)$$

Fact:

$$y_0 = 0, y_1 = \sin\left(\frac{2\pi f_n}{R}\right) \text{ [seed values]}$$

$$y_N = \alpha y_n - y_{n-2} \quad \text{for } n > 2$$

where

$$\alpha = 2\cos\left(\frac{2\pi f_n}{R}\right)$$

Ex: Discretized sin fnc with $f = 5\text{hz}$ and sampling rate $R = 10\text{hz}$

- finding recurrence coefficient...

$$\alpha = 2\cos\left(\frac{2\pi f_n}{R}\right) = 2\cos\left(\frac{2\pi 5}{10}\right) = 1.9021 \text{ [radianmode]}$$

- finding seed values...

$$y_0 = 0$$

$$y_1 = \sin\left(\frac{2\pi f_n}{R}\right) = \sin\left(\frac{2\pi 5}{10}\right) = 0.3090$$

- next 2 values of discretized sine...

$$y_2 = \alpha y_1 = y_0$$

$$y_3 = \alpha y_2 - y_1$$

Lookup Table for Sin Function

- pre-compute an array of sin function values from 0 to 2π
- To compute $\sin(x)$
 - 1) use 2π periodically to find x' in range $0 \leq x' \leq 2\pi$
 - 2) find index i in $A[i] \leq x' \leq A[i+1]$
 - 3) linearly interpolate between the adjacent samples

Ex:

$R = 10 \text{ Hz}$

lookup-table from slides

0	1	2	3	4	5	6	7	8	9	10
0	0.5878	0.9511	0.9511	0.5878	0	-0.5878	-0.9511	-0.9511	-0.5878	0

- lookup table values

$$y_n = \sin\left(\frac{2\pi f_n}{R}\right) \text{ [1 cycle so } f = 1]$$

Q. approx. $\sin(3.95)$ using lookup table:

1. fractional index into table : $y_n = \sin\left(\frac{2\pi f_n}{R}\right)$

$$3.95 = \frac{2\pi n}{R} = \frac{2\pi n}{10} = 6.2866$$

2. linearly interpolate table values